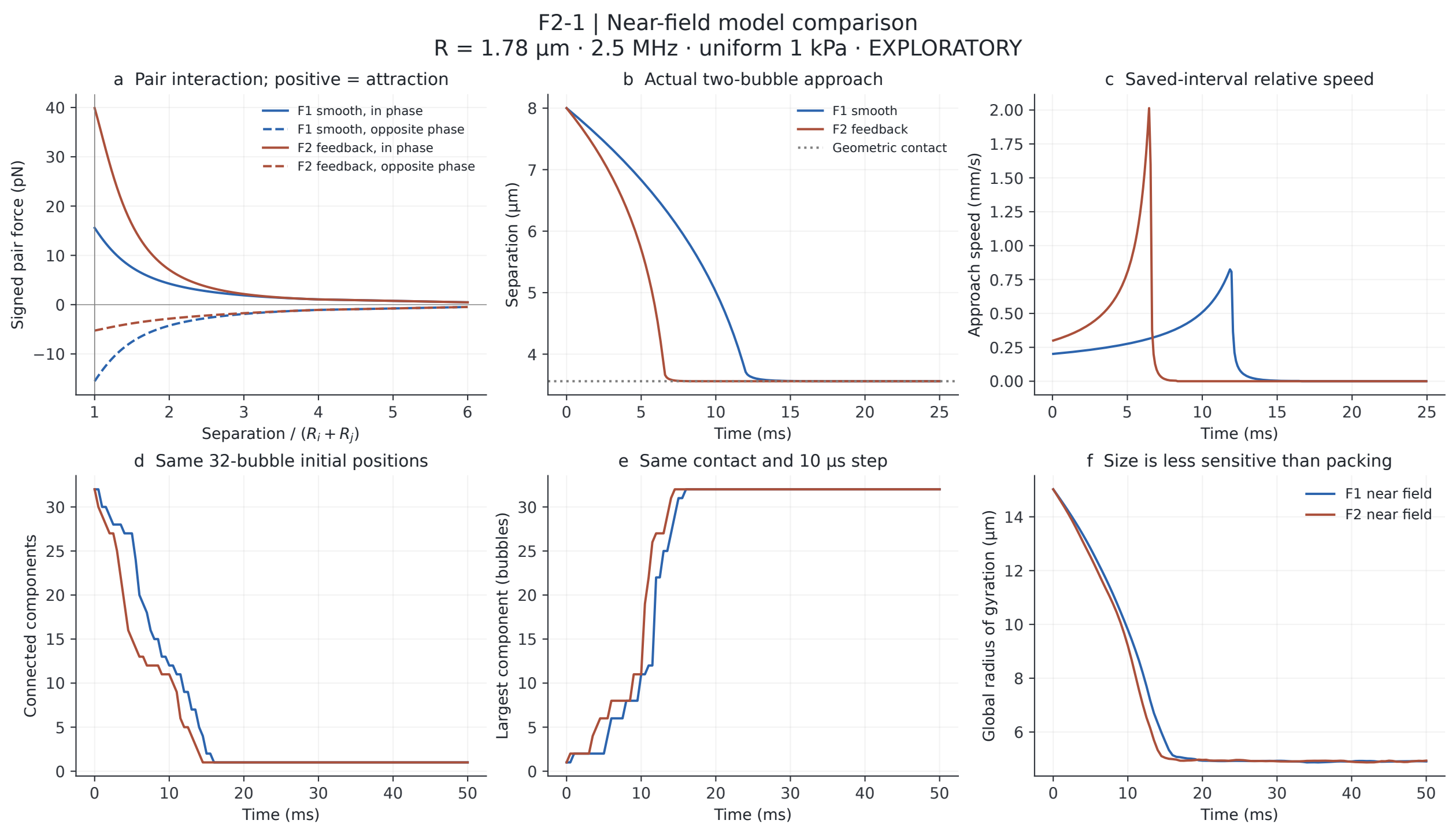
F2-1 | Near-field model comparison
R = 1.78 µm · 2.5 MHz · uniform 1 kPa · EXPLORATORY
a Pair interaction; positive = attraction
F1 smooth, in phase
F1 smooth, opposite phase
F2 feedback, in phase
F2 feedback, opposite phase
Signed pair force (pN)
Separation / $(R_i + R_j)$
b Actual two-bubble approach
F1 smooth
F2 feedback
Geometric contact
Separation (µm)
Time (ms)
c Saved-interval relative speed
Approach speed (mm/s)
Time (ms)
d Same 32-bubble initial positions
Connected components
Time (ms)
e Same contact and 10 µs step
Largest component (bubbles)
Time (ms)
f Size is less sensitive than packing
F1 near field
F2 near field
Global radius of gyration (µm)
Time (ms)